Antimicrobial Resistance in 2019' published in 2022 in the journal *The Lancet* revealed that India has one of the most worrying antibiotic resistance trends in Asia. Another study conducted by the Indian Council of Medical Research (ICMR) between January 1 and December 31, 2021 only corroborates the observation. Its estimates show that resistance towards broad-spectrum antibiotics, a group of drugs that can be used for both gram-positive and gram-negative bacteria, two categories that bacteria are commonly classified into on the basis of their cell wall structure, is rising; according to Dr Kamini Walia, who led the study, this rise may be up to 10 per cent a year.

Enter 'superbugs' or multi-drug resistant bacteria that commonly available medicines cannot treat. Particularly carbapenems, a class of antibiotic agents used to treat complicated bacterial infections. "Resistance to carbapenems," says Dr Purva Mathur, a pathologist with the All India Institute of Medical Sciences (AIIMS) who contributed to the aforementioned ICMR study, "is a huge concern as it is a powerful medicine for ICU patients and is used to treat pneumonia and septicemia. Resistance has gone up from 40 to 70 per cent." A study conducted by the Jai Prakash Narayan Trauma Centre at AIIMS last year reached similar conclusions when it found an alarmingly high rate of multi-resistance in gram negative bacteria, and again particularly against carbapenems, within their own hospital. The study noted that the numbers have been rising for the past five years.

Resistance to one drug has a cascading effect on other medicines. *Klebsiella pneumoniae*, which is one of the most common causes of hospital-acquired diseases, is growing resistant to not just carbapenems but other antibiotics as well. In 2016, the bacteria was 65 per cent vulnerable to Imipenem, a carbapenem drug; in 2021, that figure was 43 per cent. As carbapenems became less effective, says Dr Mathur, doctors began using Colistin, another powerful antibiotic. Then, in 2017, Christian Medical College (CMC), Vellore, identified six cases of *Klebsiella* that had become resistant to Colistin, too.

Doctors are alarmed. "Colistin resistance is of concern as this is an important last-hope drug. If this fails, we are out of options," says Dr V Balaji, head of clinical microbiology at CMC Vellore and of the antimicrobial stewardship programme at ICMR. "In India, bacterial resistance spreads fast due to a variety of factors. If resistance has already started growing, there is little to be done except come up with a new drug—or ensure that high-risk patients don't get infected," he says. The ICMR report also warns against other bacteria that have grown resistant. (see *India on a Slippery Slope*).

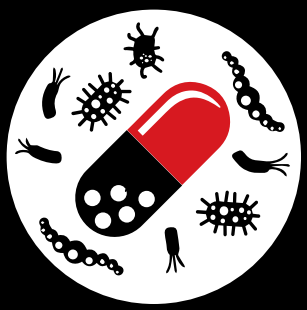
> "PEOPLE DO NOT UNDERSTAND OR RESPECT ANTIBIOTICS. THEY WANT TO GET BETTER QUICKLY, NOT CORRECTLY"
>
> **DR ABDUL GHAFUR**
> *Infectious Disease specialist, Apollo Hospital, Chennai*

ANTIBIOTICS IN YOUR FOOD?

However, the general belief that only overdosing or misusing antibiotics can put one at risk is not entirely true. When Mumbai-based banking executive Sarah Banerjee (*name changed*), 45, went for a stool test, the culture showed she had drug-resistant bacteria in her gut. She says she always completed her antibiotic courses and had never been admitted to a hospital.

Research has shown that overuse of antibiotics in the food industry could also contribute to AMR. In 2020, a study published in the journal *Diagnostic Microbiology and Infectious Disease* found that out of 65 stool samples of healthy patients at Apollo Hospital, Chennai, 51 per cent had Colistin-resistant bacteria in the gut, out of which 77 per cent were food-acquired. This is now considered the first indirect evidence of gut colonisation by food-origin Colistin-resistant bacteria. "Earlier, we were clear that either you misuse antibiotics or you acquire drug-resistant bugs in a hospital ICU where these creatures thrive on plastic-like

THE ANTIBIOTIC CRUNCH



- As of 2021, only 27 new antibiotics were being developed against priority bugs, down from 31 in 2017
- The WHO estimates that bacteria develop resistance to new drugs within two to three years of their market entry
- In 2021, the WHO said that only 12 antibiotics had been approved in the previous five years. Of these, 10 belonged to existing classes with established mechanisms of resistance
- On an average, it takes 10 years to develop a new antibiotic due to long-term safety and efficacy concerns. But only one in 15 of the existing antibiotic types reach the market; for new antibiotic classes, only one in 30 drugs make the cut